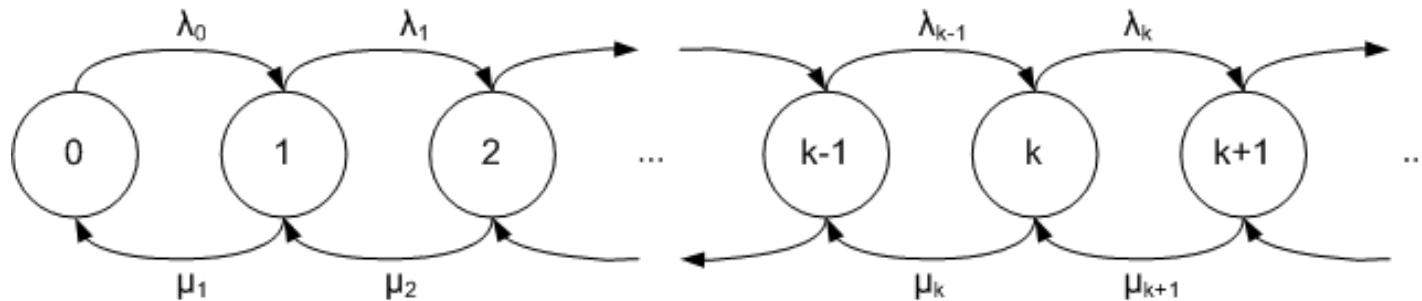


MATH 6241 – Stochastic Processes

❖ Birth-and-Death Process

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A **Birth-and-Death Processes** is a special and very important class of continuous-time Markov chains where the state represents a **count** (population size, number in system, jobs in queue, etc.), and transitions can only occur **to neighboring states**.



Birth-and-Death Processes

A **birth-and-death process** is a continuous-time Markov chain

$$\{X(t)\}_{t \geq 0}$$

with state space $\Omega = \{0, 1, 2, \dots\}$

Transitions are allowed **only between neighboring states**.

Births (to the right):

$$i \rightarrow i + 1 \text{ at rate } \lambda_i, \quad \text{for } i \geq 0.$$

Deaths (to the left):

$$i \rightarrow i - 1 \text{ at rate } \mu_i, \quad \text{for } i \geq 1.$$

Generator Matrix

The infinitesimal generator $Q = (q_{ij})$ has **tridiagonal form**:

$$Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \cdots \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 & 0 & \cdots \\ 0 & \mu_2 & -(\lambda_2 + \mu_2) & \lambda_2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Transition Graph and Reversibility

The **undirected graph** of a birth–death process is a path graph (which is a tree):



The CTMC is **time-reversible** since the state graph is a tree.

So, stationary distribution can be found by local balance equations instead of solving $\pi Q = 0$ directly.

Local Balance $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j

If $\pi = (\pi_i)$ is a stationary distribution, reversibility implies

$$\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}$$

Solving for the Stationary Distribution

$$\pi_1 = \pi_0 \frac{\lambda_0}{\mu_1},$$

$$\pi_2 = \pi_1 \frac{\lambda_1}{\mu_2} = \pi_0 \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2},$$

by induction,

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \quad \text{for } k = 1, 2, \dots$$

To make sure $\sum_{k=0}^{\infty} \pi_k = 1$

We need

$$\pi_0 = \left(\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \right)^{-1} \quad \sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} < \infty$$

Example: Continuous-Time Random Walk with reflecting boundaries.

This is a **birth–death CTMC** with **constant rates**:

$$\lambda_i = \lambda \quad (i \geq 0), \quad \mu_i = \mu \quad (i \geq 1).$$

So all births occur at rate λ , all deaths at rate μ .

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} = \pi_0 \left(\frac{\lambda}{\mu} \right)^k \quad \text{for } k = 1, 2, \dots$$

$$\pi_0 = \left(\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \right)^{-1} = \left(\sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu} \right)^k \right)^{-1} = \left(\frac{1}{1 - \frac{\lambda}{\mu}} \right)^{-1} = 1 - \frac{\lambda}{\mu}$$

$$\lambda < \mu$$

Example. Yule Process (Pure Birth CTMC)

The **Yule process** models population growth where **each individual reproduces independently** at a constant rate λ . There are no deaths.

Let X_t be the population size at time t .

If $X_t = i$, then, $i \rightarrow i + 1$ at rate $i\lambda$.

This is a birth–death process with $\lambda_i = i\lambda$, and $\mu_i = 0$.

Because there are: no deaths, and birth rates grow with i , the process drifts to infinity with probability 1.

Even though there is no stationary distribution, we can compute the distribution at finite time t .

Assume the process starts from a **single individual**: $X_0 = 1$

Let $P_{1,t} = \mathbb{P}(X_t = j \mid X_0 = 1)$

The forward equations are

$$P'_{1,j}(t) = -\lambda j P_{1,j}(t) + \lambda(j-1)P_{1,j-1}(t) \quad \text{for } j = 1, 2, 3, \dots$$

Initial condition:

$$P_{1,1}(0) = 1, \quad P_{1,j}(0) = 0 \quad (j \geq 2).$$

Solving this system (or verifying directly), we obtain

$$P_{1,j}(t) = e^{-\lambda t} (1 - e^{-\lambda t})^{j-1}$$

This is a **geometric distribution** on $(1, 2, 3, \dots)$ with parameter $p = e^{-\lambda t}$

General Initial Population

If the process starts from $X_0 = i \geq 1$, then

$$P_{i,j}(t) = \binom{j-1}{i-1} e^{-i\lambda t} (1 - e^{-\lambda t})^{j-i} \quad \text{for } j \geq i$$

This is a **negative binomial distribution**.

Example. Academic Support Center (Finite-Server Queue)

An academic support center has N **tutors**.

- Students arrive according to a **Poisson process** with rate λ .
- Each tutor serves one student at a time.
- Each service time is **exponential(μ)**, independent of everything else.
- If all N tutors are busy when a student arrives, the student **leaves immediately** (no waiting).

Let

X_t = number of busy tutors at time t .

Goal: **Find the stationary distribution of X_t .**

The state space is finite: $\Omega = \{0, 1, 2, \dots, N\}$

- $X_t = i$: exactly i tutors are busy.
- Capacity constraint: $X_t \leq N$.

This is a **loss system** (no queue).

Birth rates (arrivals)

If $i < N$, a student arrival increases the number of busy tutors by one:

$$i \rightarrow i + 1 \text{ at rate } \lambda.$$

If $i = N$, arrivals are blocked and ignored.

$$\lambda_i = \begin{cases} \lambda, & i = 0, 1, \dots, N - 1, \\ 0, & i = N. \end{cases}$$

Death rates (service completions)

If $i > 0$, there are i busy tutors.

- Each tutor's service time is exponential (μ).
- The time until **any one** of the i tutors finishes is

$$\min\{E_1, \dots, E_i\} \sim \text{Exponential}(i\mu).$$

Thus,

$$i \rightarrow i - 1 \text{ at rate } \mu_i = i\mu,$$

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda}{i\mu} = \pi_0 \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!}$$

$$\pi_0 = \left(\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \right)^{-1} = \left(\sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!} \right)^{-1}$$

Model	Birth rate (λ_i)	Death rate (μ_i)	Stationary?	Distribution
(CT random walk)	λ	μ	iff ($\lambda < \mu$)	Geometric
(Yule)	$i\lambda$	0	No	None
(Tutors)	λ	$i\mu$	Yes	Truncated Poisson

Table 1: Types of Birth-and-Death Processes

Type	Birth Rate λ_i	Death Rate μ_i
Pure birth	λ_i	$\mu_i = 0$
Poisson process	$\lambda_i = \lambda$	$\mu_i = 0$
Pure death	$\lambda_i = 0$	μ_i
Linear process	$\lambda_i = i\lambda, i > 0$	$\mu_i = i\mu$
Yule process	$\lambda_i = i\lambda, \lambda > 0$	$\mu_i = 0$
Linear with immigration	$\lambda_i = i\lambda + \alpha, \alpha > 0$	$\mu_i = i\mu$

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow